



UNIVERSITY OF
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System Design Review - Electrical

INNEX-1 Rover - Innova UoL

University of Leicester, UK Lunabotics Team

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Author(s)	Enio Mecaj
Authorised by	Tire
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Summary

This report details the electrical system architecture for the INNEX-1 Lunar Rover. The design features a Dual-Domain Power Topology that isolates high-current propulsion loads from sensitive computing hardware to prevent voltage brownouts.

The rover utilizes a single-board computer paired with a smart camera for edge-computing visual perception, which processes data onboard to significantly reduce bandwidth usage. Propulsion is driven by motor drivers managing four stepper DC motors in a skid-steer configuration. Safety systems include a hard-wired E-STOP and a power meter positioned upstream of the cut-off switch to preserve energy scoring data. This design prioritizes safety, reliability, and scoring efficiency while minimizing mass and complexity.

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1 Electrical Requirements

This section defines what the electrical subsystem must deliver to make sure the rover meets competition objectives. Requirements are derived from the rulebook, as well as system constraints.

Link from the rulebook:
https://uklunabotics.co.uk/Lunabotics/UKLunabotics_RuleBook_v0.3_20_10_2025.pdf

Req ID	Requirement	Reason
ER-1	Minimizing energy/power required	minimizing energy/power required, and maximizing autonomy. Rulebook p.10
ER-2	The system shall not use Ultrasonic Proximity Sensors	Use of Ultrasonic Proximity Sensors during off-world mission is not viable. Rulebook p.12
ER-3	The hardware shall not use touch sensors to sense or avoid obstacles	Lunabotics Competition prohibits the use of touch sensors. Rulebook p.12 Rulebook p.12
ER-4	All the parts that are responsible to movement must be disabled from power with one push motion on the E-stop button.	To ensure safe shutdown of robot in case of emergency Rulebook p.14
ER-5	No GPS to be installed on the robot. All systems that might have a integrated GPS must be disabled.	Teams may not use global positioning systems Rulebook p. 12
ER-6	The battery must be chosen within the limitations of the rules.	Examples of allowable components include Lithium-Ion (Li-ion), Nickel- UK Lunabotics Competition Rule Book 2026 13 Cadmium (Ni-Cad), Sealed Lead-Acid (SLA), or Nickel Metal Hydride (NiMH) batteries;

		Rulebook p. 13
ER-7	The E-STOP button must be COTS unmodified Red Button. The minimal diameter is 40mm.	. The robot shall be equipped with an E-STOP button. An unmodified “Commercial Off-The-Shelf” (COTS) red button is required. Rulebook p. 13
ER-8	There is only one E-STOP button, and it needs to be placed at the highest point of the robot.	The E-STOP button shall be placed on the highest practical location on the robot. There shall be only one E-STOP button per robot Rulebook p.13
ER-9	There can be a second battery powering the Pi-5 and the sensors if it doesn’t flow through the main robot E-STOP button.	it is acceptable to have a small battery onboard that only powers a Raspberry Pi control computer, and whose power does not flow through the main robot E-STOP button Rulebook p.13
ER- 10	The button must be clearly visible and easily accessible to the operator with no obstruction.	BS EN 60204-1 (Safety of machinery - Electrical equipment), state that an emergency stop must be clearly visible, identifiable, and readily accessible to the operator. It must be unobstructed and located so it can be actuated without the operator having to reach over, under, or around any part of the machine. Rulebook p.13
ER-11	The button must not reset unless it is intentionally reset and cannot reset itself automatically.	The reset action (e.g., twisting or pulling the mushroom button) shall only permit the machine to be restarted by a separate,

		deliberate start command (such as pressing a 'start' or 'reset' button). The e-stop must latch in UK Lunabotics Competition Rule Book 2026 14 the "stop" position until it is intentionally reset and cannot reset itself automatically. Rulebook p.13
ER-12	Power meter shall be placed at the highest point in the robot.	Power Meters / Data Loggers - devices shall be placed on the highest practical location on the robot and be easily visible. Rulebook p.14
ER-13	The robot should have only on-board power.	The robot shall provide its own onboard power. No facility power will be provided to the robot during the attempt. Rulebook p.14
ER-14	The power meter should be between the battery and the kill switch.	The power meter shall be schematically located between the battery and kill switch, so the readings are not erased if the E-STOP button is activated. Rulebook p.14
ER-15	Batteries can be charged only while attended while charging.	Batteries used in robots must be attended while charging. Chargers shall be unplugged overnight Rulebook p.15
ER-16	Battery containers are needed for safe storing, charging and transporting of Li-on batteries	Battery containers must be designed for safely storing, charging, and transporting lithium-ion batteries, or approved equivalent Rulebook p.15

ER-17	Batteries must be stored in upright containers and there shouldn't be any contact between them.	batteries cannot be in contact with each other. Rulebook p.15
ER-18	If a beacon is going to be used the system must not have been modified. Only Class I or II lasers or low powered ones are permitted	The beacon may send a signal or light beam or use a laser-based detection system which have not been modified (optics or power). Only Class I or Class II laser or low powered lasers (< 5mW) are allowed. Supporting documentation from the laser instrumentation vendor must be provided to the responsible faculty member for "eye-safe" lasers. Rulebook p.16
ER-19	The hardware shall display the electrical energy consumed by an (commercial off the shelf (COTS)) electronic data logger.	The judge will verify the electrical energy consumed. Rulebook p.17
ER-20	The arena cameras should be the least of cameras possible to reduce Bandwidth usage and increase the number of points.	Bandwidth use: The team will be scored on arena camera usage. The less cameras used, the higher point scored will be. Rulebook p.17
ER-21	No compasses (analogue or digital) are allowed in the robot	Use of Compass during off-world mission is not viable. Rulebook p.21

2 Power Architecture

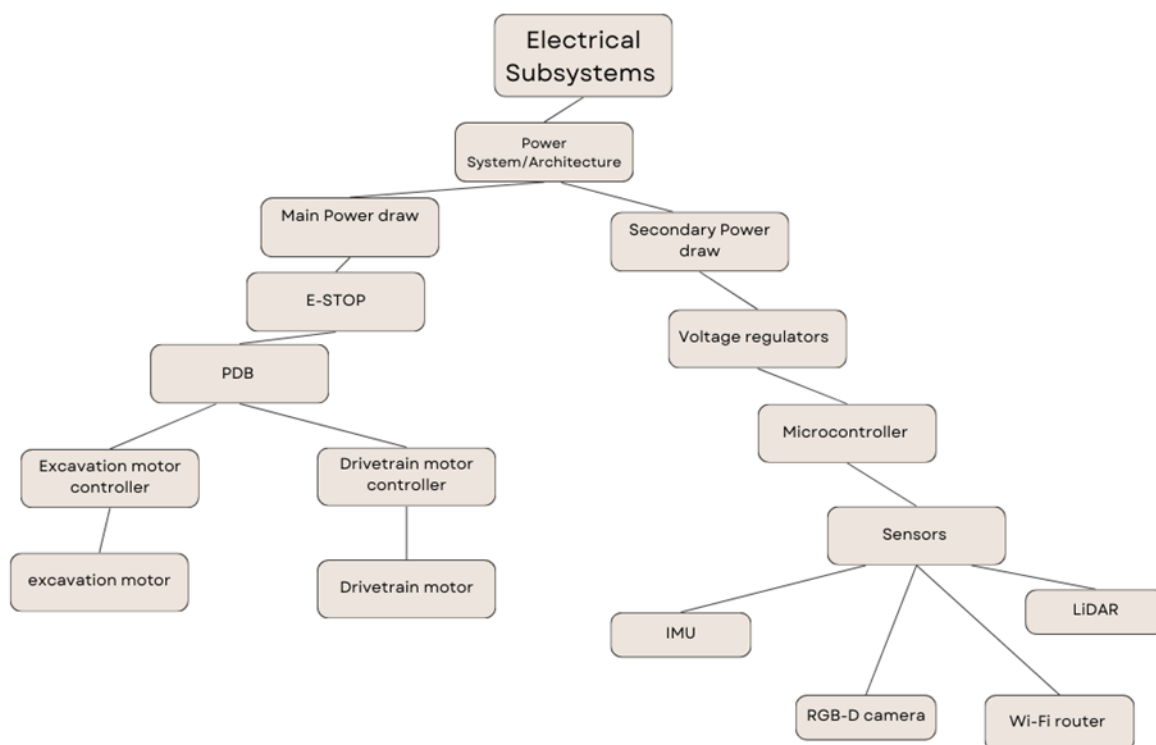
Dual-Domain Power Topology To maximize safety, reliability, and scoring efficiency, the electrical system is divided into two isolated power domains: High Power (Domain A) and Low Power (Domain B).

Field reliability studies [1] show that power and control system failures are a primary cause of UGV mission aborts. Our Dual-Domain architecture mitigates this risk by isolating the control system from motor voltage spikes.

Domain A (Motors & Excavation): Powered by a 24V battery. This bus handles all high-current inductive loads (motors and linear actuators). It is protected by a primary fuse and a master isolation switch. Crucially, the Power Meter is installed before of the E-STOP button. This ensures that in an emergency event, the robot's motion is halted immediately by physically disconnecting the battery from the motor controllers, but the power meter remains active to preserve energy consumption data for scoring.

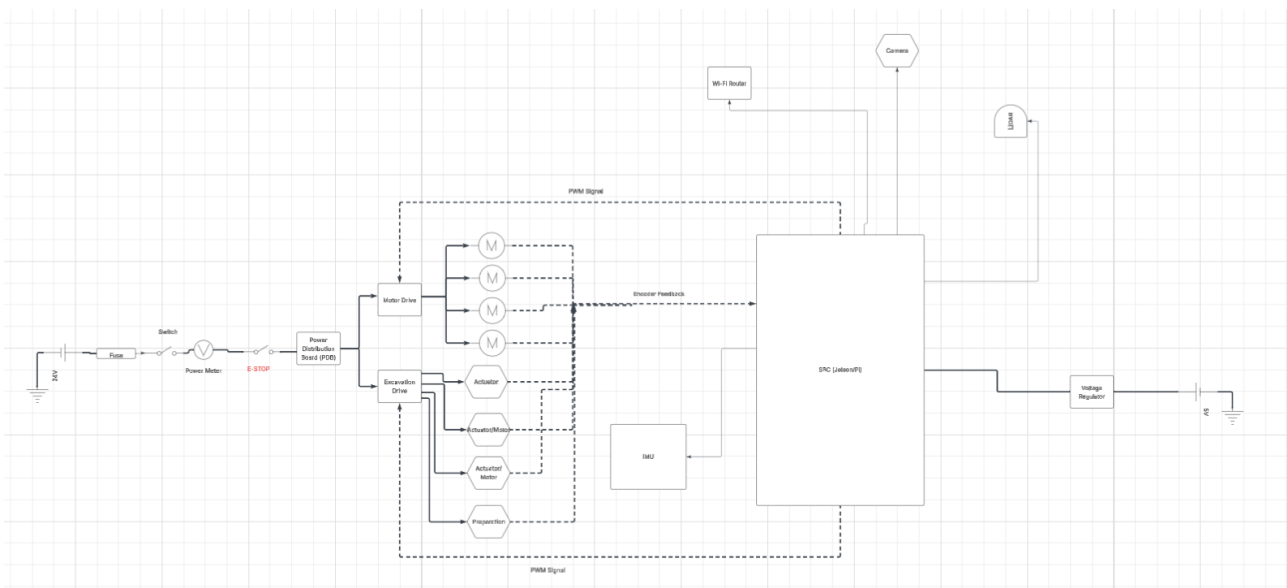
Domain B (Computing & Sensors): Powered by an auxiliary 3S LiPo. This bus powers the Single Board Computer (Raspberry Pi 5), LiDAR, cameras, and Wi-Fi networking. This domain bypasses the main E-STOP. This architecture prevents voltage drop caused by motor stalls from rebooting the computer and allows the team to receive telemetry/sensor data even if the robot is E-Stopped.

2.1 Power Architecture Diagram

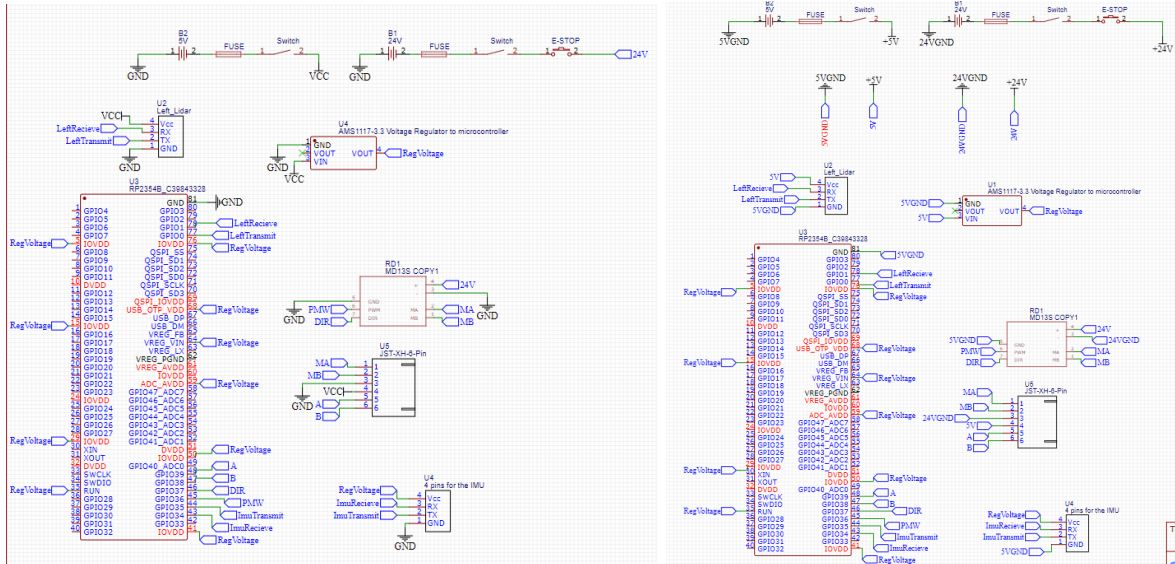


This diagram shows the dual-power system for the INNEX-1 Rover's Electrical system. Separating the high-power and the low power draws for reliable systems. There is the main power domain drawing from a 22.2V, 10,000mAh, 30C battery and the secondary power drawing from a 11.1V, 5600mAh battery. The main power passes through an E-STOP switch and a Power Distribution Board (PDB) to supply the high current loads, which include the drivetrain motors (drawing 9A per motor, so a total of 36A) and the excavation motor drawing 1.3A. the voltage will be supplied evenly from the PDB to each set of motor controllers. The secondary power branch uses a voltage regulator to condition power to 5V, supplying the low power components such as the microcontroller and the sensors. The sensors include the RGB-D camera (drawing 1.5A), the LiDAR (0.1A) and the IMU (0.1A). The microcontroller used on the above diagram is Raspberry Pi5 as the Jetson Orin nano has different power requirements.

3 Prototype Wiring Schematic



Wiring Schematic in Easy EDA:



4 Component list and suppliers

Component name	Link to model	Quantity	Price:	Notes	Sub-system
LiDAR	54.010012 Okdo Lidar Module with Bracket Development Kit for LiDAR_LD06 Raspberry Pi SBC RS	1	£48.71 (not including shipping)	Mapping and obstacle detection. Suggestion: mount with dust protection and could add lens covers.	Sensors
Camera (RGB-D camera)	https://shop.luxonis.com/products/oak-d?srsltid=AfmBOor1hYN9651POX5riPHfE4wyqtrWrdUAFqG75CzTFxjDw_lj3eQ2_t	1	£250	Use strong compression or low frame rate to remain under 4000Kbps average. Send depth slices and do onboard processing.	Sensors

IMU	TransducerM AHRS 9-Axis IMU for Robotics & Autonomous Vehicles (TM151) - RobotShop SYD Dynamics TransducerM 9-Axis AHRS / IMU w/ Dual-Port Communication (TM171) - RobotShop	1	First link: £41.54 (Shipping: £13.81) Second link: £52.62 (shipping: £13.86)	Must not use the magnetometer function. Process of disabling the magnetometer must be documented for explanation to judges.	Sensors
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Main battery: 22.2V 24V	Tattu 10,000mAh 22.2v 30C 6S1P Lipo (EC5) – The Bionic Eye	1	First link: £196 Second link: £110.66	May need BMS	Power system
Secondary battery: 18V around 40C discharge (for Jetson) or a 11.1V Li-Po + boost converter	3350mAh 18.5V 5S 35C Supersport Pro LiPo Battery - Overlander Batteries SIGP 3S Lipo Battery 11.1V 5600mAh 70C RC Battery with XT60 Connector for RC Vehicles Car Truck Tank Boat Racing Hobby: Amazon.co.uk: Toys & Games https://www.3dxr.co.uk/electronics-c78/power-management-c91/voltage-regulators-becc-c101/hobbywing-5a-ubec-5v-6v-7-4v-2-8s-v2-air-p5267	1	First link: £62.97 Second link: £37.99 Third link: £10.99	The second link is a 11.1V LiPo battery and a step-down voltage converter.	Power system
Motor controllers	13Amp 6V-30V DC Motor Driver Sabertooth Dual 2x32A 6V-24V Regenerative Motor Driver - RobotShop (for the drive motors) RoboClaw 2x15A, 6-34VDC Regenerative Motor Controller - RobotShop (for excavation)	5	First link: £15 Second link: £28 Third link: £114.14 (estimated shipping - £13.78) Fourth link:	Might need to investigate other drivers as the DC motors and	Controls System

			£92.26(estimated shipping - £13.78)	excavation motor have high torque. The third link is quite expensive, but it provides the continuous current that the chosen motors pull.	
Drivetrain Motor	GR-WM4-ENC 24V DC 37rpm Worm Gear Motor with Encoder – Gimson Robotics	4	£55		Power system
Excavation Motor	60MM Brushed Planetary Gear DC Motor 12V 24V CW/CCW High Power High Torque 60PG60S Shaft 12mm Keyway Motor - AliExpress 13	1	£59.99		Power system
Onboard computer/micro controller. Currently choosing between a Jetson nano or a Raspberry Pi5	Raspberry Pi 5 - The Pi Hut 945-13766-0005-000 NVIDIA Jetson Orin Nano Super Developer Kit Development Kit RS	1	£139.20 for the Pi5 (This is with the highest memory 16GB) £259.01 For the Jetson Orin Nano Super	Jetson does provide on board real-time processing so raw footage isn't required from cameras which can help use	Controls system

				with bandwidth.	
Voltage regulators (Linear)	https://uk.rs-online.com/web/p/voltage-regulators/2520795	2	£9.29 £10	May need an LDO (Low dropout regulator) because it has low noise therefore no interference with our sensors.	Power system
Buck converter	8 Pack Mini MP1584EN DC-DC Buck Converter Adjustable Power Supply Module 24V to 12V 9V 5V 3V: Amazon.co.uk: Business, Industry & Science				
Dual-band router (2.4 GHz + 5 GHz)	https://store.glinet.com/products/slate-plus-gigabit-travel-router-gla1300/?utm_source=website&utm_medium=productpage	1		Must disable 2.4GHz and set the 2.4GHz channel to 20MHz.	Communications
Distribution board	WUPP Boat Fuse Block, Waterproof Fuse Panel with LED Warning Indicator Damp-Proof Cover - 12 Circuits with Negative Bus Fuse Box Holder for Car Marine RV Truck DC 12-24V, Fuses Included : Amazon.co.uk: Business, Industry & Science	1	£36.19		Power system
BMS (Battery Management System)	CIJIAOPOYU Lifepo4 Lithium BMS JKBMS 4S 5S 6S 7S 8S 17S 20S 24S 12V 24V 48V Active BMS 40A 60A 100A 200A 300A(JK-BD4A24S10P) : Amazon.co.uk: Business, Industry & Science (for main battery)	1 for the main battery and another for	First link: £99.50 Second link: £7		

	iHaospace 5S Lithium Battery Protection Circuit Board, 18V 21V 30A BMS, Motherboard Type, with Cable : Amazon.co.uk: Computers & Accessories (for the secondary battery)	the secondary battery			
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5 E-STOP Implementation

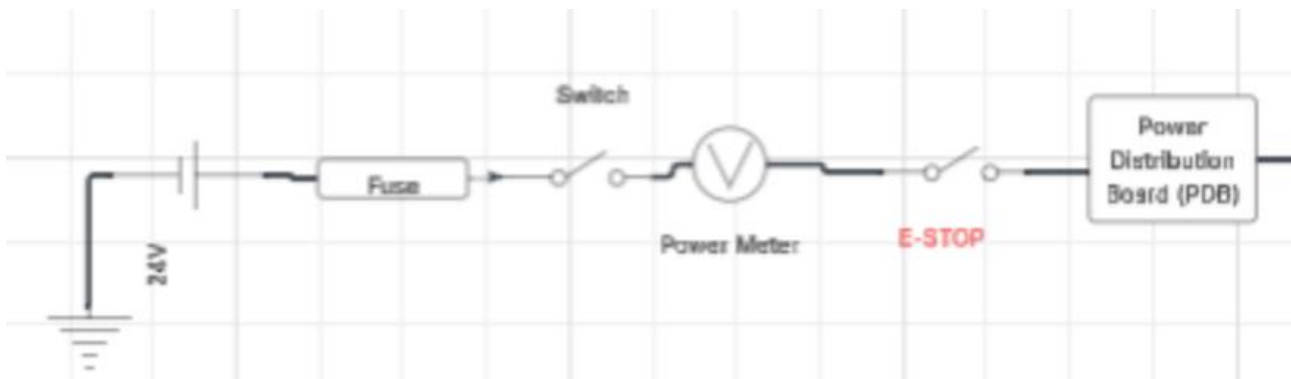
Compliance: UK Lunabotics Rules state that the E-STOP system utilizes a Red Mushroom Push-Button (Latching, Twist-to-Release) with a minimum diameter of 40mm (shown below)



Hard-Wired Safety: The E-STOP is wired in series with the positive rail of the High Power Domain (A).

Pressing the button physically breaks the connection between the Main Battery and the Power Distribution Board (PDB). This instantly cuts power to all Motor Drivers and Actuators, bringing the robot to an immediate mechanical halt.

The Compute Domain (B) remains powered via its independent battery. This ensures the Raspberry Pi and sensors stay online to log the "E-STOP Event" and allow for post-incident data analysis without losing the wireless connection to Mission Control. Diagram for the E-STOP:



6 Software Interface Summary

Motor Controller signal type –PWM

LiDAR model- [54.010012 | Okdo Lidar Module with Bracket Development Kit for LiDAR_LD06 Raspberry Pi SBC | RS](#)

Driver- LiDAR should work with both with the drivers being available on GitHub

Encoder Type A/B

Telemetry Bandwidth 230400 baud rate for one lidar 2.3×10^5 bits/s per lidar 0.23Mb/s

The OAK-D Camera has a compression module on it, H.264, and for a typical bitrate of 2-5 Mbit/s if needed for manual control, for example. If the information received is telemetry the bandwidth usage will be 0.4Mbs.

As we haven't decided on the SBC yet we have made a comparison between the pros and cons of Pi5 vs Jetson Orin Nano.

	Orin Nano	Pi 5
Price	£186-250	£90-130
CPU	6 core arm cortex A78AE 64bit	Quad core ARM cortex A76 up to 2.3 GHz
GPU/ accelerator	Nvidia Ampere gpu 1024 CUDA cores + 32 tensor cores	VideoCore 7, no CUDA/ no deep learning accelerator
RAM	8GB LPDDR5, 128-bit wide, 68Gb/s bandwidth	4GB or 8GB LPDDR4X
Storage/ expansion	microSD + M.2 NVMe	microSD (default) possibility for NVMe via adapter for SSD storage
Ports and connectivity	Gigabit ethernet: 4x USB, 3x Dp, 2x MIPI-CSI; 40 pin GPIO; plus potential for PCIe peripheral via m.2	2x USB 3.0, 2x USB 2.0, dual micro-HDMI, PCIe Gen 2.0, 2x 4 lane MIPI for camera, 40 pin GPIO header, WIFI 6, Bluetooth 5.0, Ethernet
Power consumption/ thermal	Configurable power modes 7-25W	2.5-4W, higher with heavier loads

AI performance	67 TOPS	n/a – no official figures available
ROS2 support	in principle a more powerful, more capable platform that gives you much more flexibility than Pi 5 — especially for perception, AI, multi-camera pipelines — but not without trade-offs.	technically feasible and “within norms,” but success depends heavily on camera drivers, OS/ROS distro compatibility (kernel/V4L2, Ubuntu/raspbian choice), and what processing you expect to do on Pi vs external sensor/camera hardware.
Camera limitations	Orin Nano can capture multiple camera streams, but if you need to encode (compress) and stream or record them — especially simultaneously — you are likely to saturate CPU cores and may struggle to maintain stable frame rates, especially at high resolution / high framerate.	
General limitations	• No hardware encoder • Usb bandwidth • Thermal limitations under AI • Memory bandwidth	• Weaker CPU • No GPU • Usb bandwidth • ROS2 support is worse • Thermal limitations • No ROS2 friendly hardware encoder
Bottom line	Good for multi- camera perception, running ai models, object detection, real time autonomy but usb reliability depends on power, video encoding is an issue.	Good for lightweight autonomy where AI is externally offloaded, but not good for ai interface, multicamera perception, long duration thermal stability.

Tops - TOPS (Trillions or Tera Operations per Second) is a classic example of a simplifying metric. It tells you how many computing operations an AI chip can handle in one second

References

1. J. Carlson and R. R. Murphy, "How UGVs Physically Fail in the Field," IEEE Transactions on Robotics, vol. 21, no. 3, pp. 423-437, June 2005.

2. ISO 13850:2015, "Safety of machinery – Emergency stop function – Principles for design". International Organization for
3. UK Lunabotics Competition Rule Book 2026, Version 0.3, Oct 2025.
4. LD06 TOF LiDAR Datasheet v1.0
5. Raspberry Pi 5 Product Brief
6. J. Carlson and R. R. Murphy, "How UGVs Physically Fail in the Field," IEEE Transactions on Robotics, vol. 21, no. 3, pp. 423-437, June 2005.